Customer Shopping Behavior Analysis

SQL Queries Portfolio Report

10	Business Questions
MySQL	CTEs, Aggregations & Window Functions

Prepared for GitHub portfolio presentation

Q1. Revenue by Gender

What is the total revenue generated by male vs. female customers?

```
SELECT gender,
       SUM(purchase_amount_usd) AS Total_Revenue
FROM customer_table
GROUP BY gender;
```

Q2. Discount Users Above Average Spend

Which customers used a discount but still spent more than the average purchase amount?

```
WITH average_purchase_amount AS (
    SELECT AVG(purchase_amount_usd) AS average_purchase_amount_usd
    FROM customer_table
)
SELECT customer_table.customer,
       customer_table.purchase_amount_usd
FROM customer_table
CROSS JOIN average_purchase_amount
WHERE customer_table.purchase_amount_usd >
      average_purchase_amount.average_purchase_amount_usd
  AND customer_table.discount_applied = 'Yes';
```

Q3. Top 5 Products by Average Rating

Which are the top 5 products with the highest average review rating?

```
WITH avg_review_rating AS (
    SELECT item_purchased,
           ROUND(AVG(review_rating), 2) AS average_review_rating
    FROM customer_table
    GROUP BY item_purchased
)
SELECT item_purchased, average_review_rating
FROM avg_review_rating
ORDER BY average_review_rating DESC
LIMIT 5;
```

Q4. Standard vs. Express Shipping

Compare the average purchase amounts between Standard and Express Shipping.

```
SELECT shipping_type,
       ROUND(AVG(purchase_amount_usd), 2) AS average_purchase_amount_usd
FROM customer_table
WHERE shipping_type IN ('Express', 'Standard')
GROUP BY shipping_type;
```

Q5. Subscriber Spending Comparison

Do subscribed customers spend more? Compare their average spend and total revenue with non-subscribers.

```
SELECT subscription_status,
       COUNT(customer) AS total_customers,
       SUM(purchase_amount_usd) AS total_purchased_amount_usd,
       ROUND(AVG(purchase_amount_usd), 2) AS average_purchased_amount_usd
FROM customer_table
GROUP BY subscription_status
ORDER BY total_purchased_amount_usd DESC;
```

Q6. Products with Highest Discount Rate

Which 5 products have the highest percentage of purchases with discounts applied?

```
WITH percentage_of_purchases AS (
    SELECT item_purchased,
           ROUND(
               100.0 * SUM(CASE WHEN discount_applied = 'Yes' THEN 1 ELSE 0 END)
               / COUNT(*), 2
           ) AS discount_rate
    FROM customer_table
    GROUP BY item_purchased
)
SELECT item_purchased, discount_rate
FROM percentage_of_purchases
ORDER BY discount_rate DESC
LIMIT 5;
```

Q7. Customer Segmentation

Segment customers into New, Returning, and Loyal based on previous purchases, then count each segment.

```
WITH customer_type AS (
    SELECT CASE
               WHEN previous_purchases = 1 THEN 'New'
               WHEN previous_purchases BETWEEN 2 AND 10 THEN 'Returning'
               ELSE 'Loyal'
           END AS customer_segment
    FROM customer_table
)
SELECT customer_segment,
       COUNT(*) AS number_of_customers
FROM customer_type
GROUP BY customer_segment
ORDER BY number_of_customers DESC;
```

Q8. Top 3 Products in Each Category

What are the top 3 most purchased products within each category?

```
WITH item_counts AS (
    SELECT category,
           item_purchased,
           COUNT(*) AS total_orders
    FROM customer_table
    GROUP BY category, item_purchased
),
product_ranking AS (
    SELECT category,
           item_purchased,
           total_orders,
           ROW_NUMBER() OVER (
               PARTITION BY category
               ORDER BY total_orders DESC
           ) AS ranking
    FROM item_counts
)
SELECT ranking, category, item_purchased, total_orders
FROM product_ranking
WHERE ranking <= 3
ORDER BY category, ranking;
```

Q9. Repeat Buyers and Subscriptions

Are customers with more than 5 previous purchases also likely to subscribe?

```
SELECT subscription_status,
       COUNT(customer) AS repeat_buyers
FROM customer_table
WHERE previous_purchases > 5
GROUP BY subscription_status;
```

Q10. Revenue Contribution by Age Group

What is the revenue contribution of each age group?

```
SELECT age_group,
       SUM(purchase_amount_usd) AS total_purchase_amount_usd
FROM customer_table
GROUP BY age_group
ORDER BY total_purchase_amount_usd DESC;
```